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B032426 (B255) - ARCHITECTURE, MODEL AND ANALYSIS OF CYBER PHYSICAL SYSTEMS

Bayesian Networks

Lecture 2

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Recap from last lecture

You learned how to:

- Motivate the usage of Bayesian Network (BN) modeling formalisms in the context of dependability analysis of cyber-physical systems
- Define the BN modeling formalism
- Sketch simple BN models
- Explain some relevant BN properties
- Implement and evaluate simple BN models in automated tools

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Learning objectives of this lecture

At the end of this lecture you will be able to:

- Transform a Fault Tree (FT) into a Bayesian Network (BN) for enhanced dependability modeling and analysis
- Enrich the BN with additional elements such as non-binary events, noisy-gates, and common-mode-of-failure
- Perform BN-specific analyses and evaluate the effects of model parameters on the results
- Independently build a tree-structured BN-model to model relevant dependability attributes such as reliability and safety

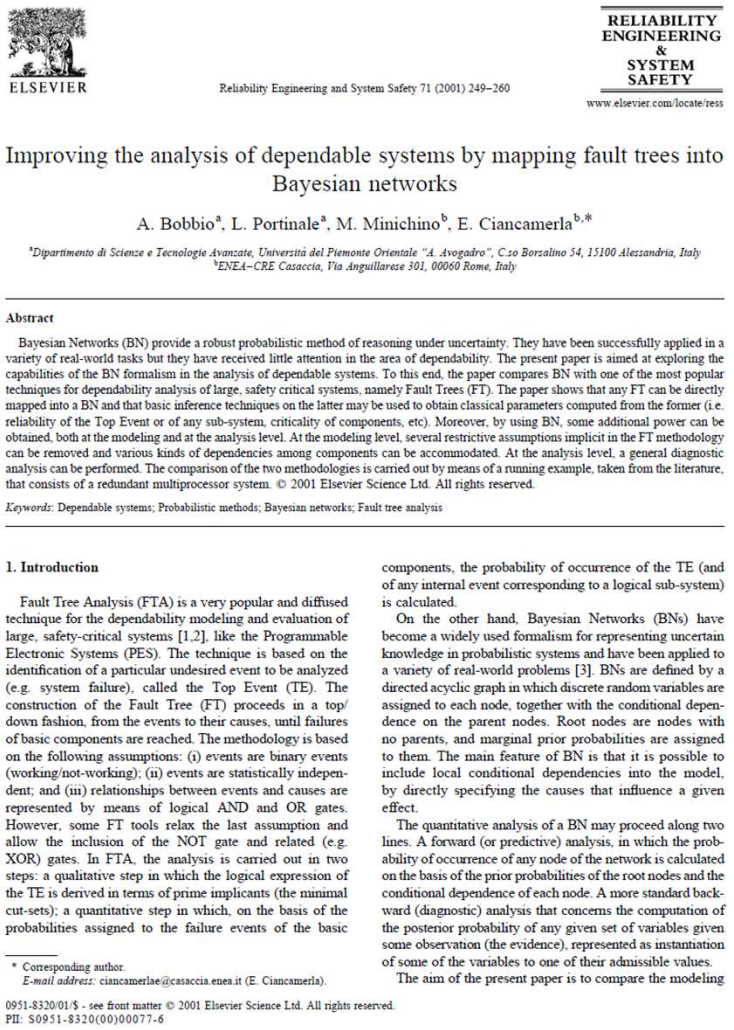
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Learning materials for this lecture

A. Bobbio, L. Portinale, M. Minichino, E. Ciancamerla, Improving the analysis of dependable systems by mapping fault trees into Bayesian networks, Reliability Engineering & System Safety, Volume 71, Issue 3, 2001, Pages 249-260, ISSN 0951-8320, https://doi.org/10.1016/S0951-8320(00)00077-6

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Improving the analysis of dependable systems by mapping fault trees into Bayesian networks

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Abstract

Bayesian Networks (BN) provide a robust probabilistic method of reasoning under uncertainty. They have been successfully applied in a variety of real-world tasks but they have received little attention in the area of dependability. The present paper is aimed at exploring the capabilities of the BN formalism in the analysis of dependable systems. To this end, the paper compares BN with one of the most popular techniques for dependability analysis of large, safety critical systems, namely Fault Trees (FT). The paper shows that any FT can be directly mapped into a BN and that basic inference techniques on the latter may be used to obtain classical parameters computed from the former (i.e. reliability of the Top Event or of any sub-system, criticality of components, etc). Moreover, by using BN, some additional power can be obtained, both at the modeling and at the analysis level. At the modeling level, several restrictive assumptions implicit in the FT methodology can be removed and various kinds of dependencies among components can be accommodated. At the analysis level, a general diagnostic analysis can be performed. The comparison of the two methodologies is carried out by means of a running example, taken from the literature, that consists of a redundant multiprocessor system. © 2001 Elsevier Science Ltd. All rights reserved.

Keywords: Dependable systems; Probabilistic methods; Bayesian networks; Fault tree analysis

1. Introduction

Fault Tree Analysis (FTA) is a very popular and diffused technique for the dependability modeling and evaluation of large, safety-critical systems [1,2], like the Programmable Electronic Systems (PES). The technique is based on the identification of a particular undesired event to be analyzed (e.g. system failure), called the Top Event (TE). The construction of the Fault Tree (FT) proceeds in a top/down fashion, from the events to their causes, until failures of basic components are reached. The methodology is based on the following assumptions: (i) events are binary events (working/not-working); (ii) events are statistically independent; and (iii) relationships between events and causes are represented by means of logical AND and OR gates. However, some FT tools relax the last assumption and allow the inclusion of the NOT gate and related (e.g. XOR) gates. In FTA, the analysis is carried out in two steps: a qualitative step in which the logical expression of the TE is derived in terms of prime implicants (the minimal cut-sets); a quantitative step in which, on the basis of the probabilities assigned to the failure events of the basic components, the probability of occurrence of the TE (and of any internal event corresponding to a logical sub-system) is calculated.

On the other hand, Bayesian Networks (BNs) have become a widely used formalism for representing uncertain knowledge in probabilistic systems and have been applied to a variety of real-world problems [3]. BNs are defined by a directed acyclic graph in which discrete random variables are assigned to each node, together with the conditional dependence on the parent nodes. Root nodes are nodes with no parents, and marginal prior probabilities are assigned to them. The main feature of BN is that it is possible to include local conditional dependencies into the model, by directly specifying the causes that influence a given effect.

The quantitative analysis of a BN may proceed along two lines. A forward (or predictive) analysis, in which the probability of occurrence of any node of the network is calculated on the basis of the prior probabilities of the root nodes and the conditional dependence of each node. A more standard backward (diagnostic) analysis that concerns the computation of the posterior probability of any given set of variables given some observation (the evidence), represented as instantiation of some of the variables to one of their admissible values.

The aim of the present paper is to compare the modeling

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Advantages of Fault Trees

- Ease of use
- Solving efficiency

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Limitations of Fault Trees

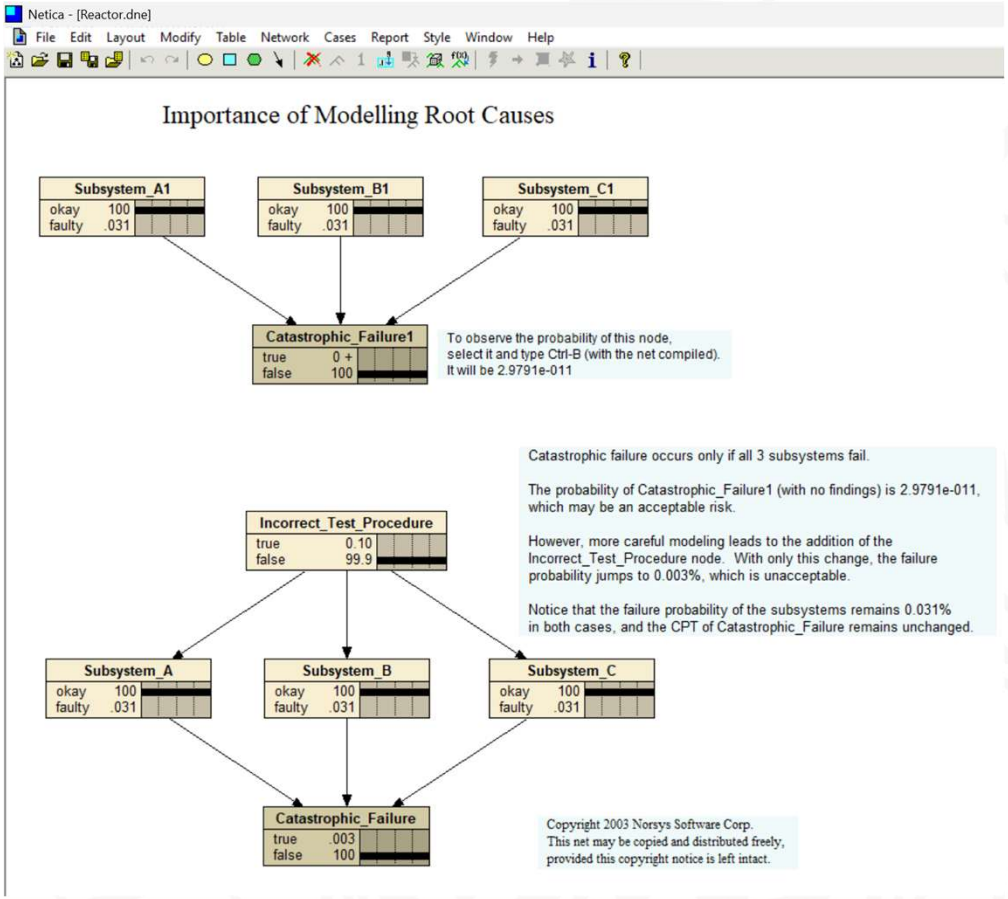
- Limited expressive power
- Impossibility to model multi-state events
- No «fuzzy» logic operators
- No common-mode failure modeling

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Netica BN example

”Reactor” model available among Netica BN examples:

C:\Netica\Netica 701\Examples



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Advantages of Petri Nets (SPN, SAN, etc.)

- Virtually unlimited expressive power
- Possibility to perform model-checking

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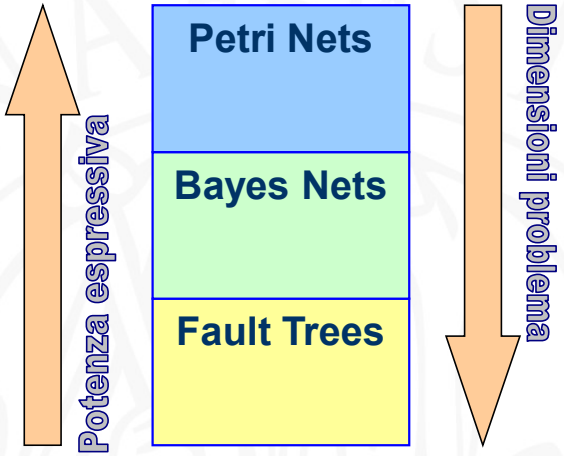
Limitations of Petri Nets (SPN, SAN, etc.)

- Difficult to develop, understand, and maintain
- State-space explosion when using analytical solvers

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Advantages of Bayesian Networks

- Good trade-off between ease of use, expressive power, and solving efficiency
- Extensions available to further increase expressive power, e.g., DBN



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Modeling options

1) Start from scratch and build the BN model using expert knowledge

2) Start from existing/pre-built FT models and translate to BN

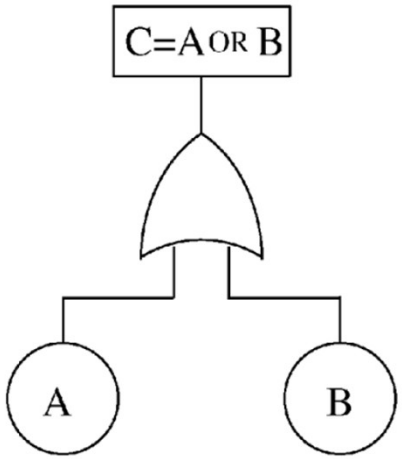
In this lecture we will show how to follow option (2)

Advantages:

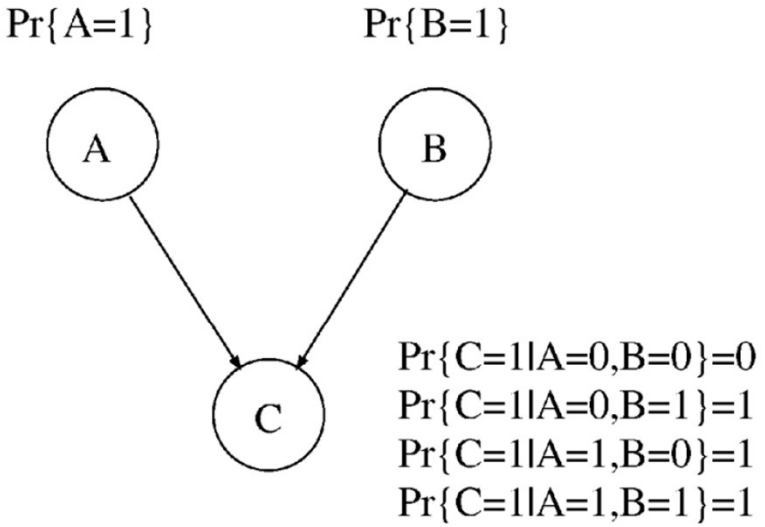
- incremental development from simpler models
- stepwise refinement and extension of proven models

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Transformation rules from FT to BN: «OR»



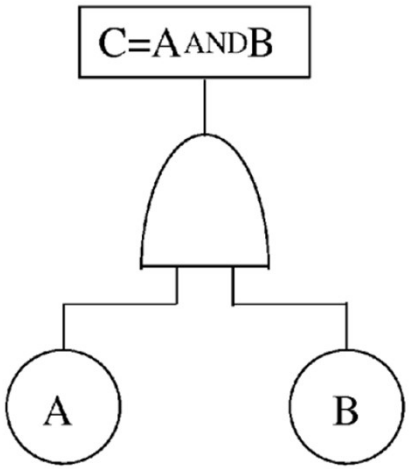
FAULT - TREE: OR Gate



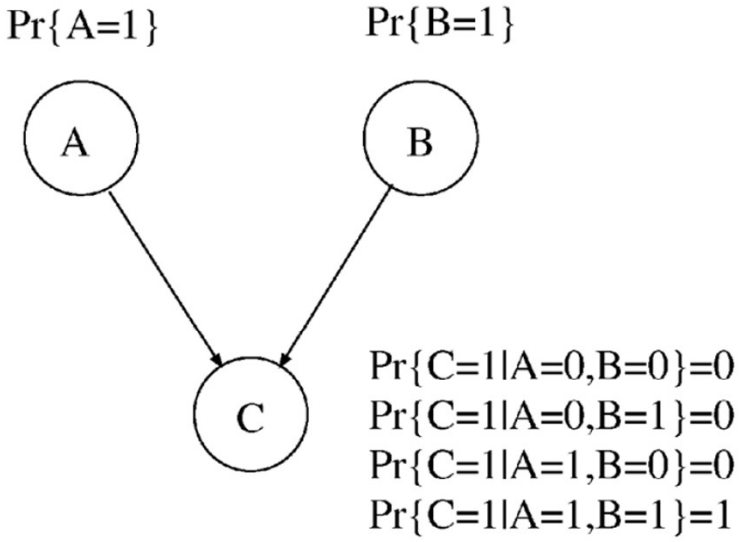
BAYESIAN NETWORK: OR Node

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Transformation rules from FT to BN: «AND»



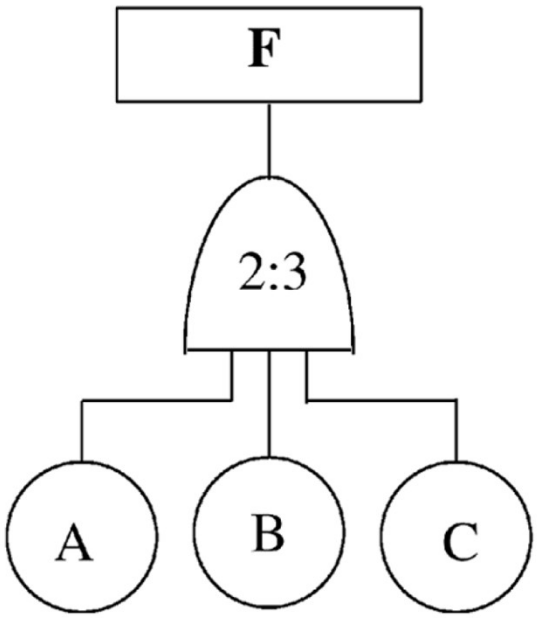
FAULT - TREE: AND Gate

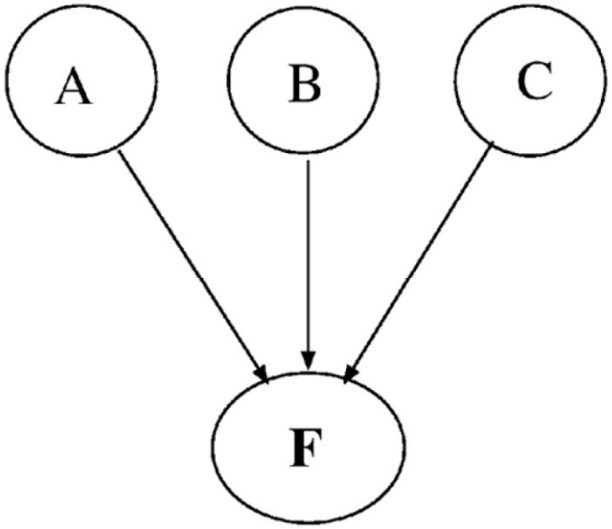


BAYESIAN NETWORK: AND Node

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Transformation rules from FT to BN: «2 out of 3»





$$\Pr\{F=1|A=0,B=0,C=0\}=0$$

$$\Pr\{F=1|A=0,B=0,C=1\}=0$$

$$\Pr\{F=1|A=0,B=1,C=0\}=0$$

$$\Pr\{F=1|A=1,B=0,C=0\}=0$$

$$\Pr\{F=1|A=0,B=1,C=1\}=1$$

$$\Pr\{F=1|A=1,B=0,C=1\}=1$$

$$\Pr\{F=1|A=1,B=1,C=0\}=1$$

$$\Pr\{F=1|A=1,B=1,C=1\}=1$$

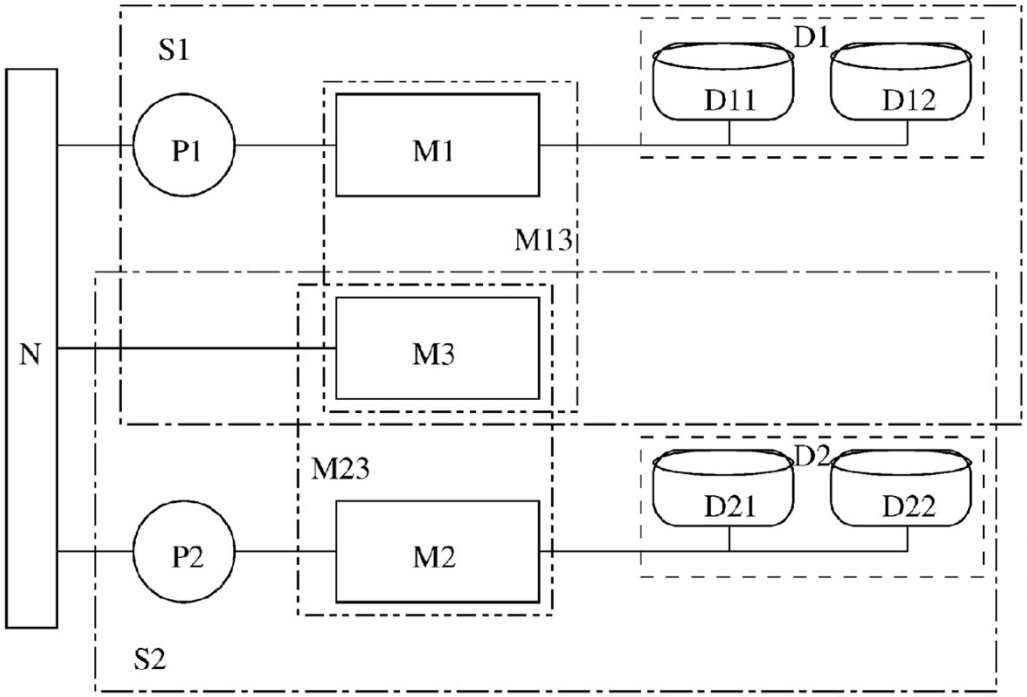
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Transformation rules from FT to BN: basic assumptions

1. events are binary events (working/not-working)
2. events are statistically independent
3. relationships between events and causes are represented by logical AND and OR gates
4. the root of the FT is the undesired Top Event (TE) to be analyzed.

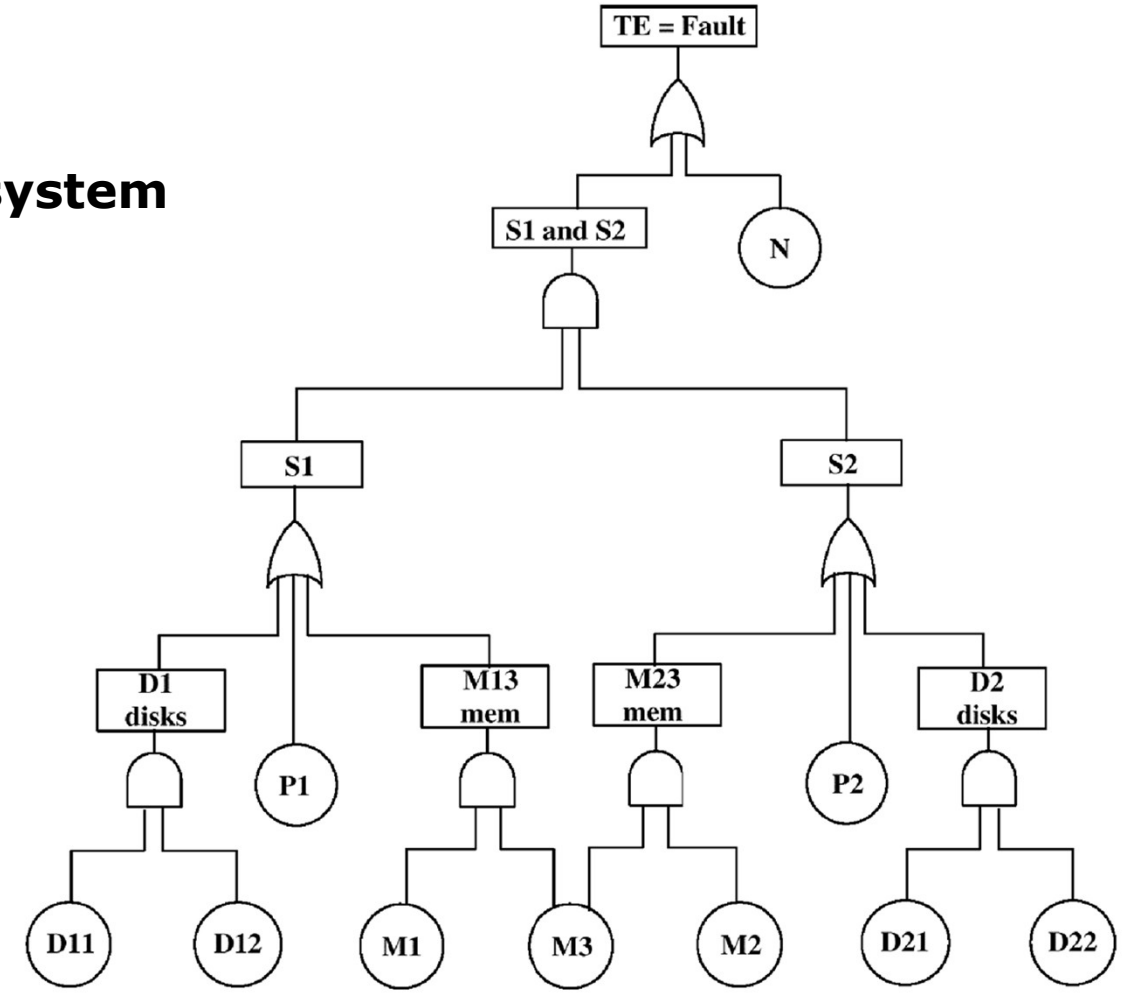
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Example: redundant multiprocessor system



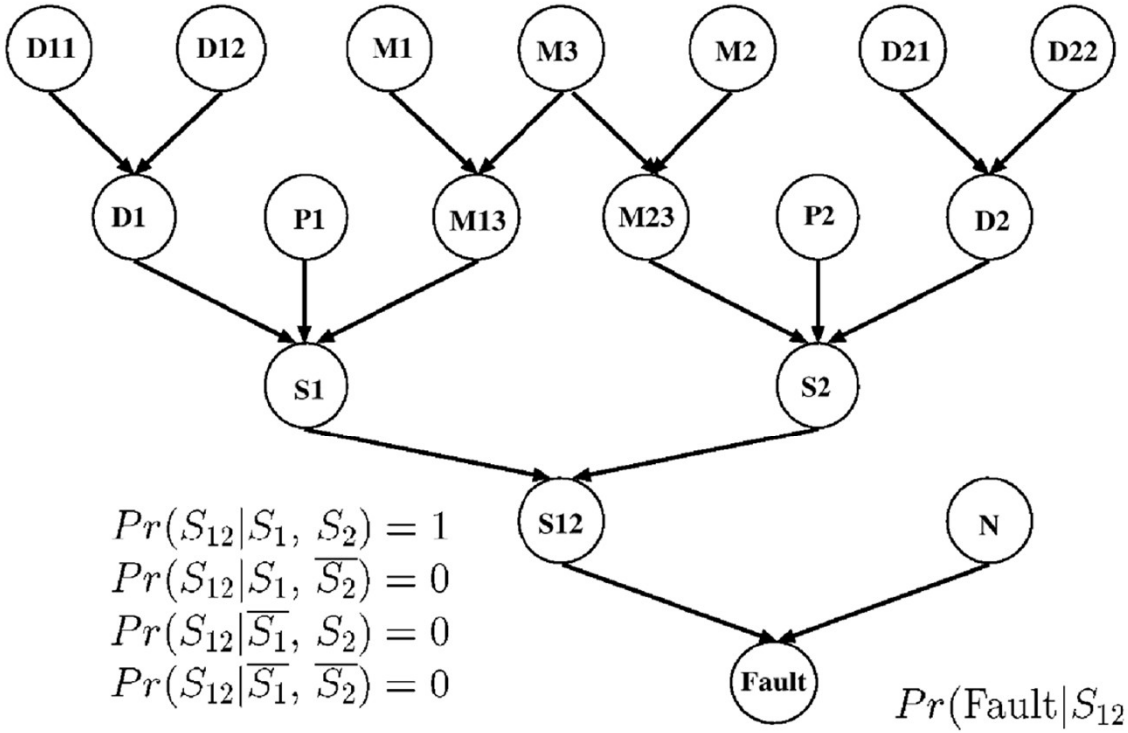
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FT model of the example redundant multiprocessor system



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BN model of the example redundant multiprocessor system



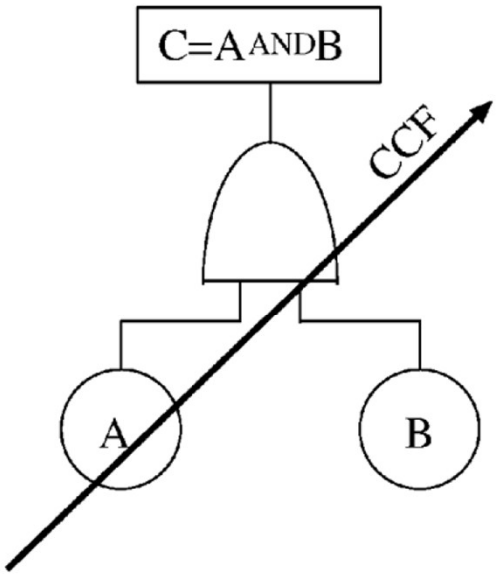
b)

$$Pr(S_{12}|S_1,\ S_2) = 1$$
$$Pr(S_{12}|S_1,\ \overline{S_2}) = 0$$
$$Pr(S_{12}|\overline{S_1},\ S_2) = 0$$
$$Pr(S_{12}|\overline{S_1},\ \overline{S_2}) = 0$$

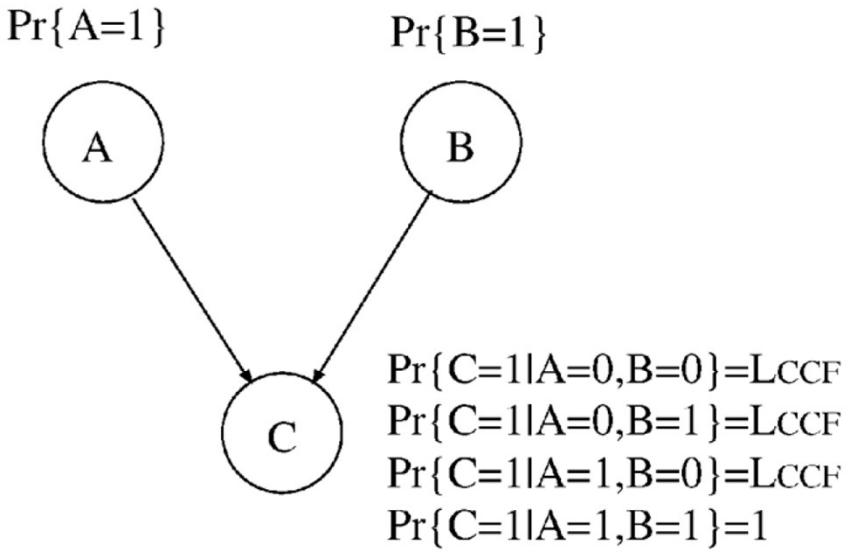
$$Pr(\text{Fault}|S_{12},\ N) = 1$$
$$Pr(\text{Fault}|S_{12},\ \overline{N}) = 1$$
$$Pr(\text{Fault}|\overline{S_{12}},\ N) = 1$$
$$Pr(\text{Fault}|\overline{S_{12}},\ \overline{N}) = 0$$

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Representation of common mode of failures in BN



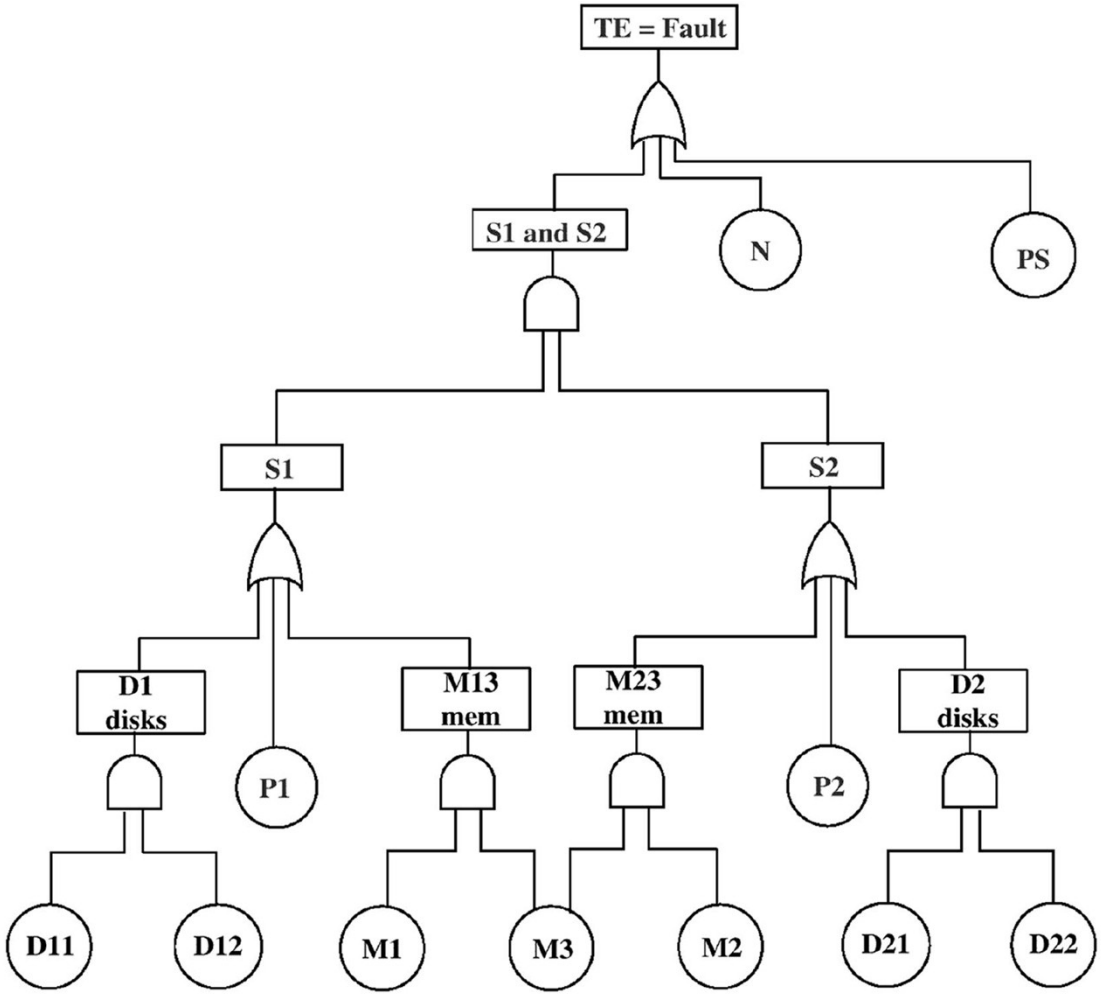
FAULT - TREE: AND Gate
With Common Cause Failures



BAYESIAN NETWORK: AND Node
With Common Cause Failures

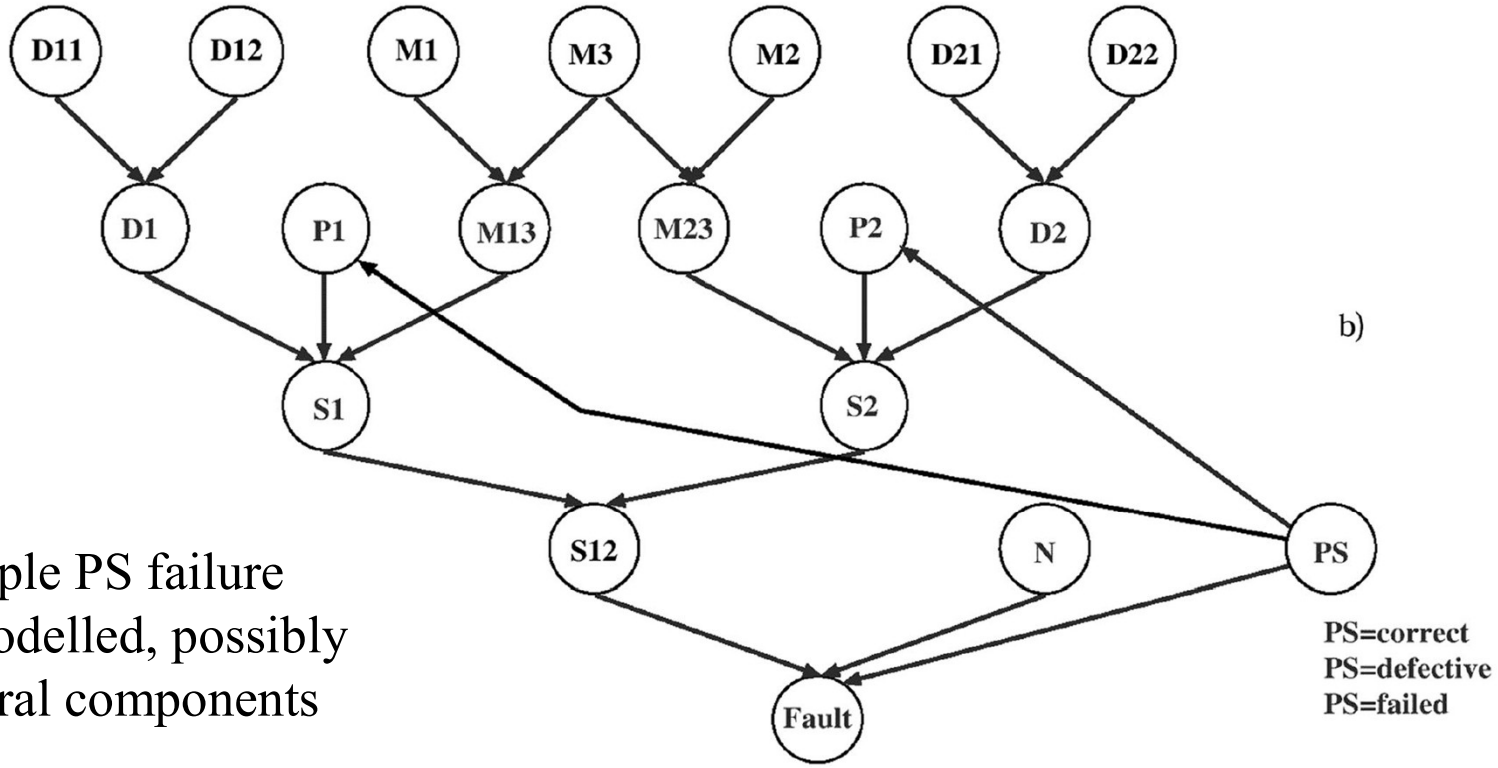
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Adding power supply: Fault Tree



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Adding power supply: Bayesian Network



With BNs, multiple PS failure modes can be modelled, possibly influencing several components

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Classroom assignment

- Transform the following Fault Tree into a Bayesian Network
- Fault-tolerant multi-processor system (P1 and P2) with two dedicated memory modules (M1 e M2) and one shared memory (M3)

```
 1   ftree t-proc-mem
 2   basic p1 exp(1/Ptime)
 3   basic p2 exp(1/Ptime)
 4   basic m1 exp(1/Mtime)
 5   basic m2 exp(1/Mtime)
 6   repeat m3 exp(1/Mtime)
 7   and mems1 m1 m3
 8   and mems2 m2 m3
 9   or pm1 p1 mems1
10   or pm2 p2 mems2
11   and top pm1 pm2
12   end
13
14   bind
15   Ptime 720
16   Mtime 2*720
17   end
18
19   cdf(t-proc-mem)
20   pqcdf(t-proc-mem)
21   end
```

MTBF

Failure Rate = 1/MTBF

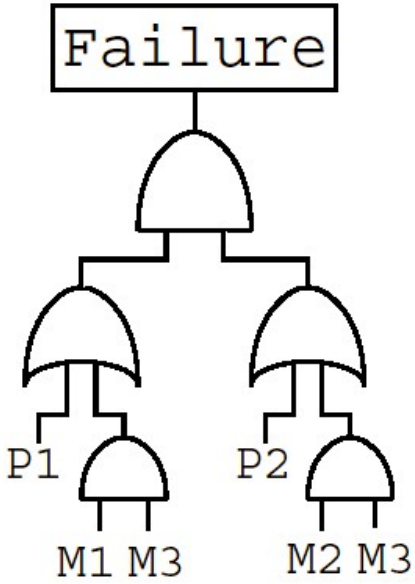
```
CDF for system t-proc-mem:

   1.0000e+00 t( 0) exp( 0.0000e+00 t)
+ -4.0000e+00 t( 0) exp(-2.0833e-03 t)
+  2.0000e+00 t( 0) exp(-2.7778e-03 t)
+  1.0000e+00 t( 0) exp(-3.4722e-03 t)
+  1.0000e+00 t( 0) exp(-4.1667e-03 t)
+ -1.0000e+00 t( 0) exp(-4.8611e-03 t)

mean: 8.7771e+02
variance: 3.5797e+05

CDF for system t-proc-mem:

[Q(p1) * Q(p2)]+
[Q(p1) * P(p2) * Q(m2) * Q(m3)]+
[P(p1) * Q(p2) * Q(m1) * Q(m3)]+
[P(p2) * Q(m1) * Q(m2) * Q(m3) * (1 − Q(p1))]
```



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Home assignment: RAID 5 storage system

- Model the following failure modes of a RAID 5 storage system with 4 units: 1) working at full performance; 2) working at degraded performance due to 1 unit failure; 3) failure.
- Consider a single unit MTBF of 200.000h. Start from a FT, translate it to a BN, and then add and model the third failure mode.
- Implement and evaluate your model in Netica.
- Prepare a few slides Powerpoint presentation about the model and results to be shown and discussed during next lecture.
- Assignment is for learning purposes only, and therefore not graded.
- Working in small groups is allowed.

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Next lecture

- Quick recap and questions about today's lecture

- Presentation of BN models developed by students

- BN modelling of real-world CPS case-study in the railway domain

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Thank you for your attention

Questions?

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